Identification of Largest 20 Environmental Startups in India

Anirul Islam

1 Search Methodology

1.1 Objective

The objective of this stage was to identify a comprehensive list of environmental startups in India that could subsequently be evaluated to identify the 20 largest startups. A reliable initial source was required to identify relevant startups, followed by the collection of additional company-level information suitable for quantitative comparison.

1.2 Selection of the Initial Data Source

The Ministry of Electronics and Information Technology (MeitY) Startup Hub was selected as the initial source for identifying environmental startups in India. The platform provides a directory of startups with information such as company names, locations, industries, and sector classifications.

The directory is available at:

https://msh.meity.gov.in/network/startup

The use of this directory provided a structured starting point for constructing the list of startups. However, the directory itself did not provide all the quantitative financial and organizational information required for the subsequent analysis.

1.3 Filtering and Identification of Startups

The startup directory provides a filtering facility based on different sectors. Two sector categories were selected for this study:

1. Renewable Energy

2. Waste Management

These categories were selected because they are directly relevant to the environmental focus of the assignment. Other available categories were not included in the initial search because they were considered less directly aligned with the scope of this study.

After applying these filters, the directory displayed 130 startup cards. These records formed the initial set of candidate environmental startups for further data collection.

1.4 Extraction of Basic Startup Information

The filtered directory page was downloaded in HTML format. A Python program was then developed to parse the saved HTML file and extract the information available in the startup cards.

The following variables were successfully extracted:

Table 1: Basic information extracted from the MeitY Startup Hub

Variable	**Description**
Startup Name	Name of the startup as displayed in the directory.
Industry	Industry or business classification displayed on the card.
State	State or territory associated with the startup.
City	City associated with the startup.
Sector Tags	Sector labels assigned to the startup in the directory.

The extracted information was organized into a structured dataset. This automated extraction process reduced the need to copy the basic details of each startup manually.

Although these variables were useful for identifying and classifying the startups, they were insufficient on their own for a quantitative comparison of company size. Therefore, additional company-level variables were collected from publicly available sources.

1.5 Collection of Additional Quantitative Information

To supplement the initial directory data, a Python program was developed to generate search URLs for individual startups. The generated links were used to search for official company websites and publicly available information relating to the following variables:

1. **Annual Revenue:** The reported annual revenue of the startup, where available.
2. **Number of Employees:** The reported number of employees, used as an indicator of organizational size.

3. **Total Funding Raised:** The total funding reported to have been raised by the startup, where available.

The official website of each startup was also sought to support company identification and facilitate the collection and verification of information.

The search URLs generated by the Python program were used to locate relevant sources. The information obtained through these searches was compiled into a structured CSV file containing the basic startup details and the additional quantitative variables.

The search process was intended to improve the coverage of company-level information beyond what was available in the original MeitY directory. The availability and precision of the collected information nevertheless depended on the disclosures made by individual startups and the information accessible from public sources.

1.6 Treatment of Missing and Approximate Information

A limitation of the data collection process was that some startups did not publicly disclose all the required financial or organizational information. In addition, some reported values were available only as estimates or within a range rather than as exact figures.

The following rules were applied when compiling the dataset:

1. **Information not publicly available:** If a required value could not be found in the available sources, the corresponding cell was left empty. Missing information was not automatically replaced with zero.

2. **Revenue reported as below INR 10 crore:** According to the information consulted on Tracxn, some startups had annual revenue reported as less than INR 10 crore. Where the exact revenue amount was not available, INR 10 crore was entered as an approximate value for these cases. Such entries should therefore be interpreted as approximations rather than exact reported revenues.

3. **Bootstrapped startups with no reported funding:** For startups identified as bootstrapped, for which no external funding was reported, total funding raised was recorded as zero. This treatment distinguishes the absence of reported external funding for these startups from cases in which funding information was unavailable.

4. **Unreported funding or revenue:** Where the available information did not establish a value, the relevant cell was kept empty rather than assigning zero or introducing an unsupported estimate.

These rules were intended to preserve the distinction between a reported zero and unavailable information. Nevertheless, the approximate revenue entries and the possibility

of incomplete public disclosures need to be considered when interpreting the resulting dataset.

1.7 Limitations of the Search Process

Several limitations arose during data collection.

1. **Limited information in the initial directory:** The MeitY Startup Hub provided basic startup information, but the initial cards did not contain all the financial and organizational variables required for comparison.
2. **Incomplete public disclosure:** Some startups did not disclose their annual revenue, employee count, or funding information publicly. Consequently, the resulting dataset contains missing values.
3. **Differences in the precision of reported values:** Some financial information was available only in approximate form or as a range. In particular, revenue values reported as below INR 10 crore were represented by INR 10 crore when an exact amount was unavailable. These values are not directly equivalent to exact revenue observations.
4. **Potential differences across information sources:** Information obtained from company websites, startup databases, and other public sources may differ in reporting period, definition, or level of detail. These differences can affect comparability across startups.
5. **Scope of the initial search:** The candidate list was constructed using the Renewable Energy and Waste Management filters in the MeitY Startup Hub. Environmental startups not classified under these categories in the directory may therefore be absent from the initial list.

Despite these limitations, the procedure established a structured initial dataset of environmental startups and supplemented the directory information with additional company-level variables. The resulting dataset provides the input for the subsequent analysis and ranking stage.

2 PCA-Based Ranking Methodology

To identify the 20 largest environmental startups among those identified through the search process, Principal Component Analysis (PCA) is employed to construct a composite ranking based on three quantitative indicators: annual revenue, number of employees, and

total funding raised. PCA is a multivariate statistical technique that transforms a set of potentially correlated variables into a smaller number of uncorrelated linear combinations, known as principal components. This approach provides a data-driven way to combine multiple indicators into a single composite score without assigning subjective weights to them in advance.

2.1 Selection of Indicators

Three indicators are considered for the ranking:

1. **Annual Revenue:** Annual revenue reflects the financial activity and business scale of a startup.
2. **Number of Employees:** The number of employees provides an indication of the startup's workforce and operational scale.
3. **Total Funding Raised:** Total funding raised represents the financial resources secured by a startup.

Other potential indicators, such as the number of products or services offered, were considered but excluded because sufficiently complete information was not available for a substantial proportion of the identified startups. The analysis therefore focuses on the three indicators for which data were collected.

2.2 Complete-Case Selection

Let N denote the total number of startups identified through the search process. The PCA analysis is restricted to startups for which observations are available for all three selected indicators.

Let x_{ij} denote the value of the jth indicator for the ith startup, where $j = 1, 2, 3$. A startup is retained only when its annual revenue, number of employees, and total funding raised are all available as valid numerical observations. Startups with missing or non-numeric values in any of these indicators are excluded from the analysis.

Missing values are not treated as zero because the absence of publicly available information does not necessarily imply that the corresponding quantity is zero. This complete-case approach ensures that all startups included in the PCA are compared using the same set of indicators.

However, restricting the analysis to complete observations may introduce selection bias if the availability of information is associated with startup size or other characteristics.

Accordingly, the resulting ranking is interpreted as applying to the startups retained for the analysis rather than necessarily representing every environmental startup in India.

2.3 Standardization of Indicators

The three indicators are measured on different numerical scales. Annual revenue and total funding raised are monetary measures, whereas the number of employees is a count. Consequently, applying PCA directly to the original variables could allow indicators with larger numerical variances to dominate the analysis.

To address this issue, each indicator is standardized using its sample mean and standard deviation. The standardized value of the jth indicator for the ith startup is defined as

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j},$$

where $\bar{x}_j$ and s_j denote the sample mean and sample standard deviation of the jth indicator, respectively.

The standardized variables have a mean of zero and a standard deviation of one. This transformation places the indicators on a comparable scale and ensures that their differing units of measurement do not directly determine their contributions to the PCA.

2.4 Principal Component Analysis

Let $\mathbf{Z}$ denote the standardized data matrix containing the three indicators for all retained startups. PCA is performed using the covariance matrix of the standardized variables:

$$\mathbf{S} = \frac{1}{n-1}\mathbf{Z}^\top\mathbf{Z},$$

where n is the number of startups retained in the complete-case dataset.

The eigenvalues and corresponding eigenvectors of $\mathbf{S}$ are obtained by solving

$$\mathbf{S}\mathbf{v}_k = \lambda_k\mathbf{v}_k,$$

where λ_k is the eigenvalue associated with the kth principal component and $\mathbf{v}_k$ is its corresponding eigenvector.

The eigenvalues are arranged in descending order:

$$\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq 0.$$

The first principal component (PC1) captures the largest possible proportion of the total variance in the standardized indicators. It is expressed as a linear combination of the three standardized variables:

$$\text{PC1}_i = a_{11} z_{i1} + a_{21} z_{i2} + a_{31} z_{i3},$$

where a_{j1} denotes the coefficient associated with the jth standardized indicator in the first principal component.

The proportion of total variance explained by the kth principal component is calculated as

$$\text{PVE}_k = \frac{\lambda_k}{\sum_{j=1}^{3} \lambda_j} \times 100\%.$$

The proportion of variance explained by PC1 is therefore

$$\text{PVE}_1 = \frac{\lambda_1}{\lambda_1 + \lambda_2 + \lambda_3} \times 100\%.$$

This measure indicates the proportion of the total variation in the three standardized indicators captured by the first principal component.

2.5 Interpretation of Principal Component Loadings

The loadings of the principal components are examined to understand the contribution of each indicator. The loading of the jth indicator on the kth principal component is defined as

$$\ell_{jk} = a_{jk} \sqrt{\lambda_k},$$

where a_{jk} is the corresponding eigenvector coefficient.

Particular attention is given to the loadings of annual revenue, number of employees, and total funding raised on PC1. If these loadings are positive and substantively interpretable, the first principal component may be regarded as a composite dimension reflecting overall startup size.

Since the signs of principal component eigenvectors are mathematically arbitrary, all loadings and scores of a component can be multiplied by -1 without changing the underlying PCA solution. For interpretability, the orientation of PC1 is chosen so that the sum of its loadings is positive.

If the resulting PC1 loadings have mixed signs, the component is not automatically interpreted as an overall startup-size measure. Its interpretation must instead be based on the direction and relative contribution of the three indicators.

2.6 Construction of the Composite Ranking Score

After examining the loadings and establishing whether PC1 provides a meaningful representation of overall startup size, the first principal component score is used as the composite ranking score.

For startup i, the score is defined as

$$C_i = \mathrm{PC1}_i = \sum_{j=1}^{3} a_{j1} z_{ij},$$

where C_i denotes the PCA-based composite score, z_{ij} is the standardized value of the jth indicator, and a_{j1} is its coefficient in PC1.

The composite score combines the three indicators according to coefficients determined by the observed covariance structure of the data. Thus, the relative contributions of the indicators are derived from the PCA rather than specified through manually chosen weights.

A higher score indicates a greater position in the positive direction of PC1. When the component has positive and interpretable loadings for all three indicators, a higher score can be interpreted as indicating a relatively larger startup according to the selected measures.

2.7 Final Ranking and Selection of the Top 20

All retained startups are ordered in descending order of their PC1 scores. The startup with the highest score is assigned rank 1, the startup with the second-highest score is assigned rank 2, and the procedure continues until all retained startups have been ranked.

The 20 startups with the highest composite scores are selected as the final PCA-based top 20. Their rankings are interpreted relative to the startups included in the complete-case analysis and the three indicators used to construct the composite score.

2.8 Methodological Considerations and Limitations

Although PCA provides a data-driven method for combining the selected indicators, its first principal component maximizes explained variance rather than startup size directly. Consequently, the interpretation of PC1 as an overall startup-size index depends on the observed loadings and their substantive meaning.

Furthermore, revenue, employment, and funding may be correlated, and the resulting component reflects the joint variation among these indicators. The PCA-based ranking should therefore be understood as a composite ranking based on the selected measures rather than a comprehensive assessment of startup quality, innovation, environmental impact, or long-term success.

The analysis also depends on the availability and comparability of publicly reported information. Differences in reporting periods, monetary units, and the precision of reported values may affect the results. Any approximate observations should be identified appropriately, and the resulting ranking should be interpreted with these limitations in mind.

Overall, PCA provides a systematic approach to ranking the environmental startups with complete observations by combining annual revenue, number of employees, and total funding raised into a single composite score.

3 Results and Discussion

3.1 Analytical Sample and Data Availability

The PCA-based ranking is performed using only those environmental startups for which complete numerical information is available for all three selected indicators: annual revenue, number of employees, and total funding raised. Startups with missing or non-numeric values in any of these indicators are excluded from the analysis. This ensures that every startup included in the ranking is evaluated using the same set of variables and that the comparison is not affected by differences in the number of available indicators.

The complete-case restriction is particularly important because the absence of publicly available information does not necessarily indicate that a startup has zero revenue, no employees, or no funding. Accordingly, missing observations are not treated as zero. The resulting ranking therefore represents the relative positions of startups with complete information for the three selected indicators, rather than a definitive ranking of every environmental startup in India.

3.2 Correlation Analysis of the Selected Indicators

The relationships among annual revenue, number of employees, and total funding raised are examined using their correlation matrix. The correlation matrix, presented in Table 2, provides an initial understanding of the degree to which these indicators vary together across the retained startups.

Table 2: Correlation Matrix of the Startup Indicators

Indicator	**Annual Revenue**	**Number of Employees**	**Total Funding Raised**
Annual Revenue	1.000	0.506	0.343
Number of Employees	0.506	1.000	0.442
Total Funding Raised	0.343	0.442	1.000

Positive correlations indicate that higher values of one indicator tend to be associated with higher values of another, whereas negative correlations indicate an inverse association. The strength and direction of these relationships are relevant to PCA because the method summarizes the joint variation among the indicators.

The correlation structure also helps explain why PCA is appropriate for this analysis. Rather than considering the three indicators independently, PCA constructs linear combinations that summarize their variation. However, correlation alone does not establish that one indicator causes changes in another or that the indicators measure every dimension of startup size.

3.3 Eigenvalues and Variance Explained

The eigenvalues and the corresponding proportions of variance explained by the principal components are reported in Table 3. Since the analysis uses three standardized indicators, three principal components are obtained.

Table 3: Eigenvalues and Variance Explained by the Principal Components

Component	**Eigenvalue**	**Variance Explained (%)**	**Cumulative Variance (%)**
PC1	1.8938	62.14	62.14
PC2	0.6741	22.12	84.26
PC3	0.4798	15.74	100.00

The first principal component (PC1) captures the largest possible proportion of the total variance in the standardized data. The second and third principal components account for the remaining variation in mutually orthogonal directions. The cumulative percentage

of variance indicates how much of the total variation is captured by the components considered together.

For the purpose of ranking the startups, PC1 is used as the composite score after examining its loadings and assessing whether it can reasonably be interpreted as a common dimension of startup size. The percentage of variance explained by PC1 indicates how effectively this component summarizes variation in the three indicators. Nevertheless, a high proportion of explained variance alone does not establish that the component represents startup size; its loadings must also be interpreted.

3.4 Interpretation of Principal Component Loadings

The loadings of annual revenue, number of employees, and total funding raised on the three principal components are presented in Table 4. These loadings describe how the standardized indicators contribute to the principal components and help establish the substantive interpretation of PC1.

Table 4: Principal Component Loadings of the Startup Indicators

Indicator	**PC1**	**PC2**	**PC3**
Annual Revenue	0.7904	-0.4914	-0.3868
Number of Employees	0.8445	-0.1122	0.5386
Total Funding Raised	0.7455	0.6481	-0.2001

Particular attention is given to the signs and magnitudes of the PC1 loadings. If all three loadings are positive and substantively meaningful, PC1 can be interpreted as a composite dimension that generally increases with revenue, employment, and funding. In this case, startups with higher PC1 scores tend to exhibit higher values across the selected indicators, supporting the use of PC1 as an overall startup-size index.

If the loadings differ in sign, the interpretation requires greater caution because the component may represent a contrast between indicators rather than a common measure of size. The signs of principal component loadings are arbitrary up to a simultaneous reversal of the component and its scores; therefore, their interpretation depends on the relative signs and substantive meaning of the loadings.

The loading structure is consequently used to justify the interpretation of the composite score rather than assuming that PC1 automatically represents startup size.

3.5 PCA-Based Ranking of Environmental Startups

After examining the correlation structure, variance explained, and principal component loadings, the retained startups are ranked in descending order of their PC1 scores. The resulting ranking is presented in Table 5.

The PC1 score combines the standardized values of annual revenue, number of employees, and total funding raised according to the coefficients determined by PCA. A higher score indicates a greater position in the positive direction of PC1. When the component has positive and interpretable loadings for all three indicators, higher scores can be understood as indicating a relatively larger startup according to the selected measures.

The 20 startups with the highest PC1 scores are selected as the final top 20. This ranking provides a systematic comparison of the retained startups using a common composite score rather than relying on any single indicator or on manually assigned weights.

It is important to distinguish this ranking from a universal assessment of startup performance. A startup may have substantial revenue but relatively few employees, or it may have raised considerable funding without generating equivalent revenue. PCA combines the observed information across the three indicators, and the resulting rank reflects their joint contribution to the first principal component.

3.6 Scope and Limitations of the Ranking

The results must be interpreted in light of the data availability and the selected indicators. First, the complete-case restriction excludes startups for which one or more of the three variables are unavailable. If the availability of information is associated with startup size, the retained sample may not be fully representative of all startups identified through the search process.

Second, publicly reported financial and employment information may differ in reporting period, definition, and precision. Approximate values should therefore be identified appropriately, and comparisons involving such observations should be interpreted cautiously.

Third, PCA maximizes the variance explained by a linear combination of the standardized indicators; it does not directly optimize startup size, environmental performance, innovation, or long-term success. The ranking should therefore be understood specifically as a PCA-based ranking of environmental startups according to annual revenue, number of employees, and total funding raised.

Overall, the PCA approach provides a transparent, data-driven procedure for constructing a composite ranking of the environmental startups with complete observations for all three indicators. The final top 20 should be interpreted within the scope of the available data and the variables included in the analysis.

3.7 Final Ranking: Top 20 Environmental Startups

Table 5 presents the final ranking of the 20 environmental startups with the highest PC1 scores among those retained in the complete-case analysis. The table reports each startup's rank, name, and corresponding PC1 score.

Table 5: Top 20 Environmental Startups Based on the PCA Method

Rank	**Startup Name**	**PC1 Score**
1	Repos energy india pvt ltd	7.1509
2	Raptee Energy Private Limited	5.2228
3	COMFY FOOTWEAR	4.1809
4	Sundarban Chemicals private limited	2.7379
5	MiniMines Cleantech Solutions Pvt. Ltd.	1.1202
6	RAHEJA SOLAR FOOD PROCESSING PRIVATE LIMITED	0.5839
7	MobiGarage (Fuevation Technology Solutions Pvt Ltd)	0.5389
8	Cropcoin Technologies Private Limited	0.3440
9	Adiabatic Technologies Private Limited	0.3288
10	Carrus Mobility Private Limited	0.2257
11	Cairovision Pvt Ltd	0.2225
12	Svabag Labs Private Limited	0.2218
13	Ceir Mobility Private Limited	0.2165
14	RapidEVChargE	0.2158
15	PEMMAN VIDYUT PRIVATE LIMITED	0.2158
16	SRI GAYATHRI METAL	0.0550
17	Green Aero Propulsion Private Limited	-0.0552
18	ScrapJI	-0.0576
19	PRADCO	-0.0898
20	Admighty Foundation	-0.0900